

Text as Data

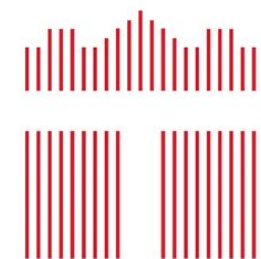
W10 Case: Text Classification and Regression

Yen-Chieh Liao 廖彥傑

2026-11-13

Fall 2026 (115-1) | National Taiwan University (GIND)

NTLDEV 7191 | Fri. 16:30-18:20



**National
Taiwan
University**
國立臺灣大學

Electoral reform and targeted oversight

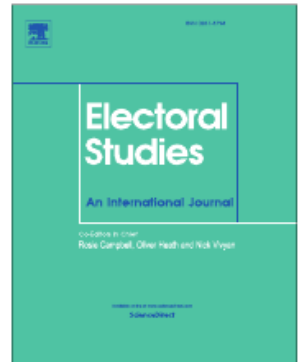
Electoral Studies 99 (2026) 103026



Contents lists available at ScienceDirect

Electoral Studies

journal homepage: www.elsevier.com/locate/electstud



Electoral systems and geographically targeted oversight: Evidence from the Taiwan Legislative Yuan[☆]

Yen-Chieh Liao^{a,1,*} , Li Tang^{b,2}

^a *University of Birmingham, Birmingham, B15 2TT, UK*

^b *Department of Economics, University of Birmingham, Birmingham, B15 2TT, UK*



Did Taiwan's electoral reform change geographical targeting in written questions?

Fine-tuning a model for a research construct

Pre-trained Chinese language model + labeled legislation

Stage	Number of legislative texts
Training	5,772
Development	722
Final source-domain test	722

7,216 texts: 2,004 particularistic / 5,212 non-particularistic.

The paper compares Chinese BERT, CKIP ALBERT, and HFL MacBERT.

A new text genre needs a new validation sample

Training domain: legislation. Application domain: written questions.

Evaluation	Reported BERT result
722 held-out legislative texts	Macro F1: 0.95; accuracy: 0.96
About 400 human-coded questions	GTQ F1: 0.88
Agreement on those questions	Pearson correlation: 0.82

Correlation, F1, and accuracy are different quantities.

Predictions become a legislator-year outcome

Question text → predicted GTQ label → legislator-year summary

$\text{GTQ percentage} = 100 \times \text{targeted questions} / \text{all questions}$

- Corpus: 63,748 questions from 402 district legislators.
- At the reported 0.95 confidence threshold: 8,992 GTQs.
- Join summaries to political and municipal characteristics.

The regression observation is a **legislator-year**, not a question.

Regression tests the political argument

Outcome: **GTQ percentage** for each legislator-year.

- Post-reform indicator and a time trend.
- **Post-reform × time** interaction.
- Legislator controls and **municipality fixed effects**.
- Standard errors clustered by **municipality and year**.

Later models examine interactions with municipal socioeconomic conditions.

Reading the published regression table

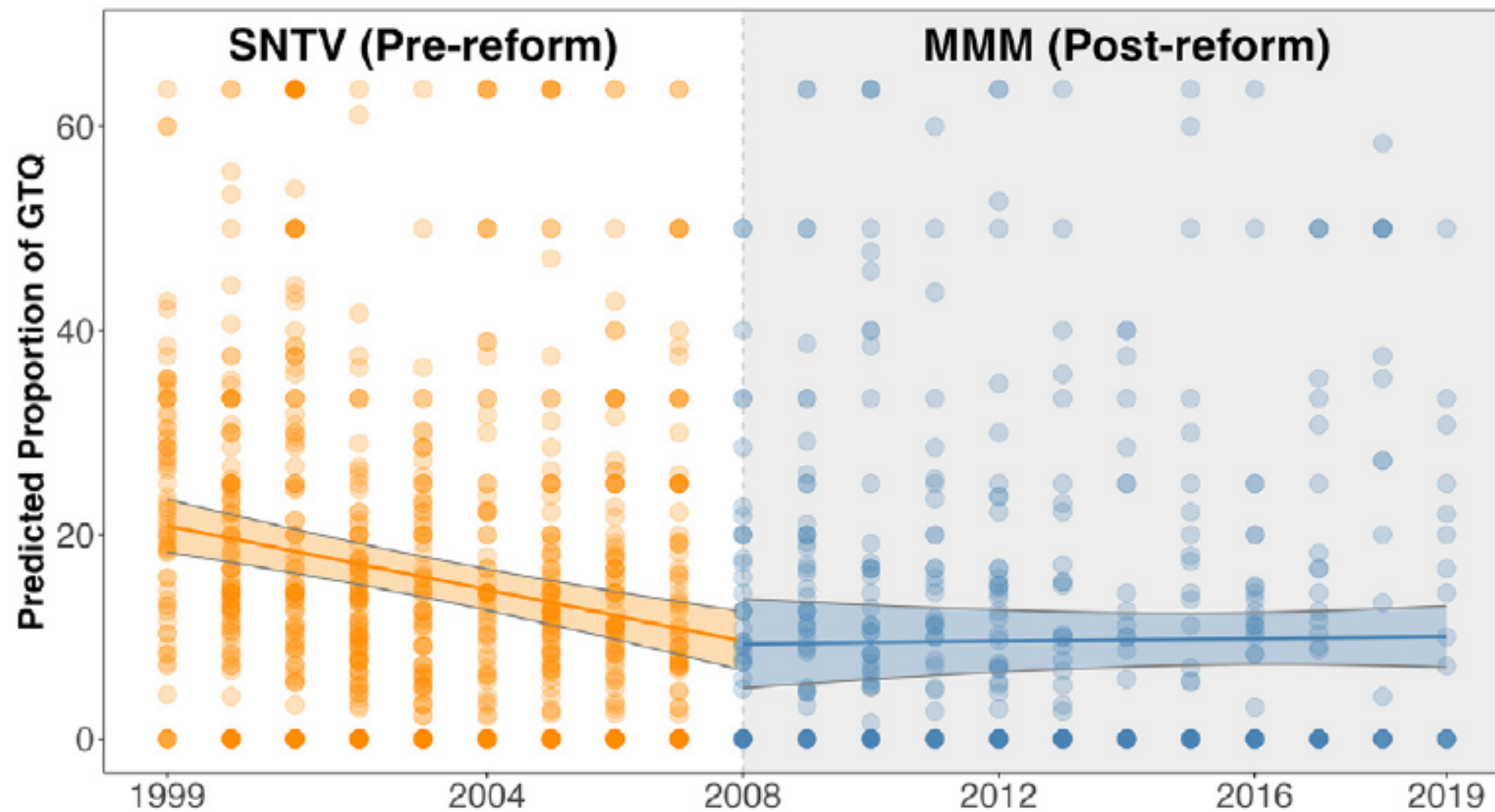
Table 1
Electoral system change and geographically targeted questions.

	Full District Legislators		Trimmed Samples	
	Model 1 (Baseline)	Model 2 (Full Controls)	Model 3 (Baseline)	Model 4 (Full Controls)
Post-reform (MMM)	− 9.716** (4.227)	− 10.396** (4.878)	− 10.076** (4.637)	− 12.057** (5.554)
Year	− 1.413*** (0.311)	− 1.325*** (0.302)	− 1.311*** (0.360)	− 1.232*** (0.354)
Post-reform × Year	1.401*** (0.413)	1.303*** (0.396)	1.339*** (0.454)	1.289*** (0.443)
Municipality FE	✓	✓	✓	✓
Observations	1,583	1,582	1,306	1,306
R ²	0.120	0.124	0.117	0.120

Negative post-reform coefficient; positive interaction with time.

Liao and Tang (2026), Table 1, p. 6; coefficient panel excerpt, CC BY 4.0. Clustered SE in parentheses; ** p < .05, *** p < .01. [DOI](#).

The estimated pattern changes over time



The figure presents **Model 4 predictions**, not classification performance.

Two kinds of validity

Measurement question	Research-design question
Do labels match the construct?	Does the comparison test the hypothesis?
Do errors vary by period or group?	What else changed around the reform?
How does the threshold affect GTQ rates?	How do trends and controls affect estimates?

Good prediction does not, by itself, establish a causal effect.